

# flora

electronic garden

AMAZE.

Explorer ★ Award Winner ★  
A MAZE Berlin 2025 🐌

Judit Castells - Interaction Design  
Daniela Guevara - Interaction Design  
Paula Jaime - Art Direction  
Mariana Pachón - Programming & Music



## Concept



Flora is in crisis. This electronic garden has been cursed. The flowers no longer bloom, the snails have lost their shells, the river has run dry, and the entire landscape lies still, as if everything had shut down. It seems that all is lost, but with help, this ecosystem will shine again. This electronic garden needs to be cared for, someone to contemplate it and figure out how to reintegrate each species back into the cycle.

Flora is an interactive installation inspired by a natural ecosystem. It arises from the need to challenge individualistic dynamics and our lack of awareness of certain natural symbioses. The installation invites players to reconnect with forgotten skills, such as play and collaboration. It serves as a space for exploration, listening, and becoming part of a greater cycle. Players uncover the relationships between organisms and help them communicate and thrive in harmony.

This circular board features three-dimensional objects that simulate life through electronic elements, functioning in unity and cooperation. Movements are powered by four Arduinos that communicate with six sensors and nine actuators. For example, planting a seed is simulated by inserting a moisture sensor into damp soil. When the sensor detects increased moisture, the code activates the water pump and other actuators, simulating the flow of a river and the blooming of flowers. Additionally, when the snail moves, a card beneath its body triggers a response as it passes over the RFID reader, illuminating magical stones that unlock various abilities.

Playtime connects us with the freedom of being in a space where rules are shaped by unique experiences. Sometimes, there may be no purpose at all. Play for the joy of playing, and allow yourself to be surprised by the sound, colors, and textures.

If you want to know more about Flora here's a presentation video:  
<https://www.youtube.com/watch?v=s-QiC-kuT5c&t=3s>



## Playtime

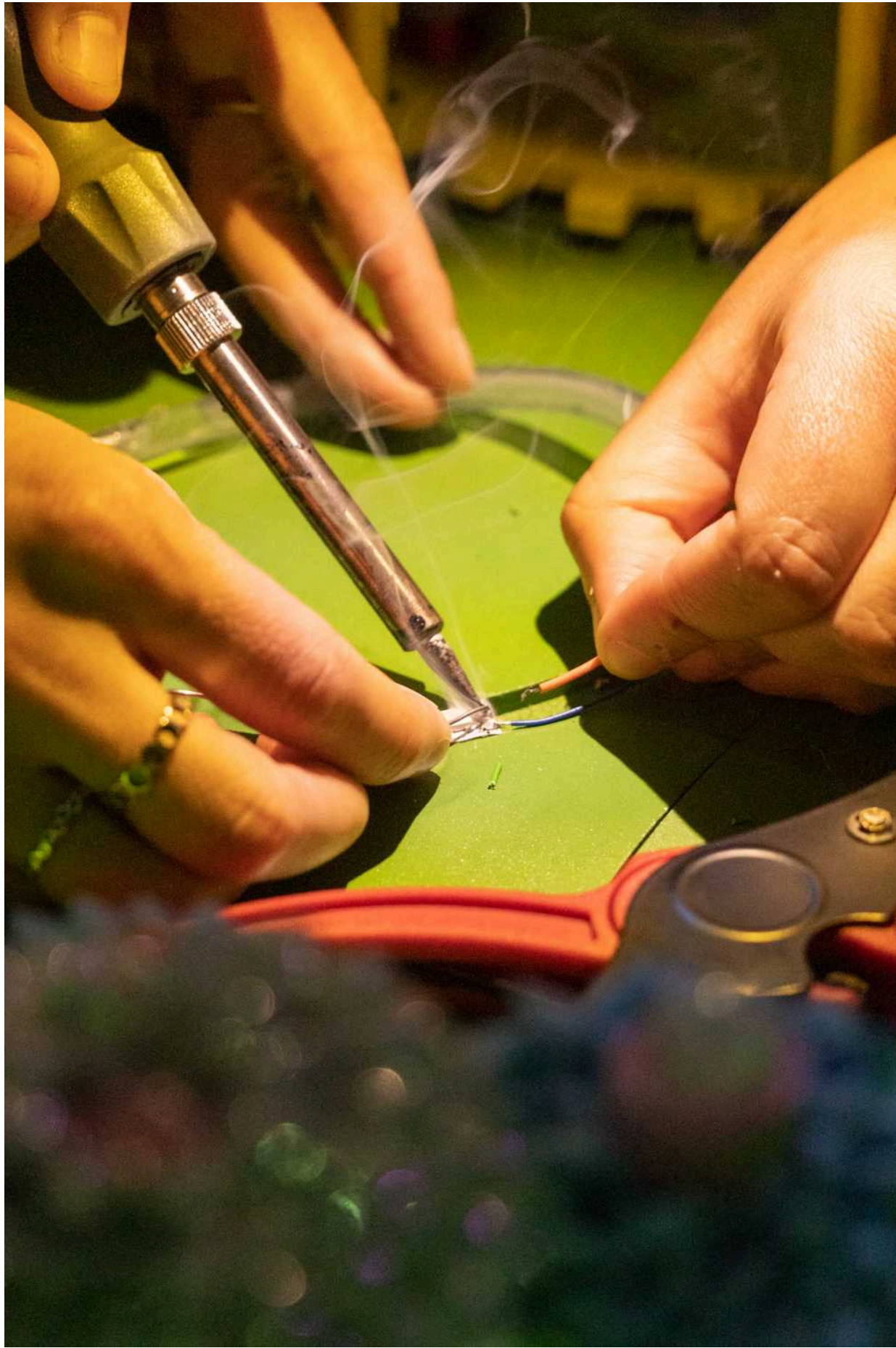
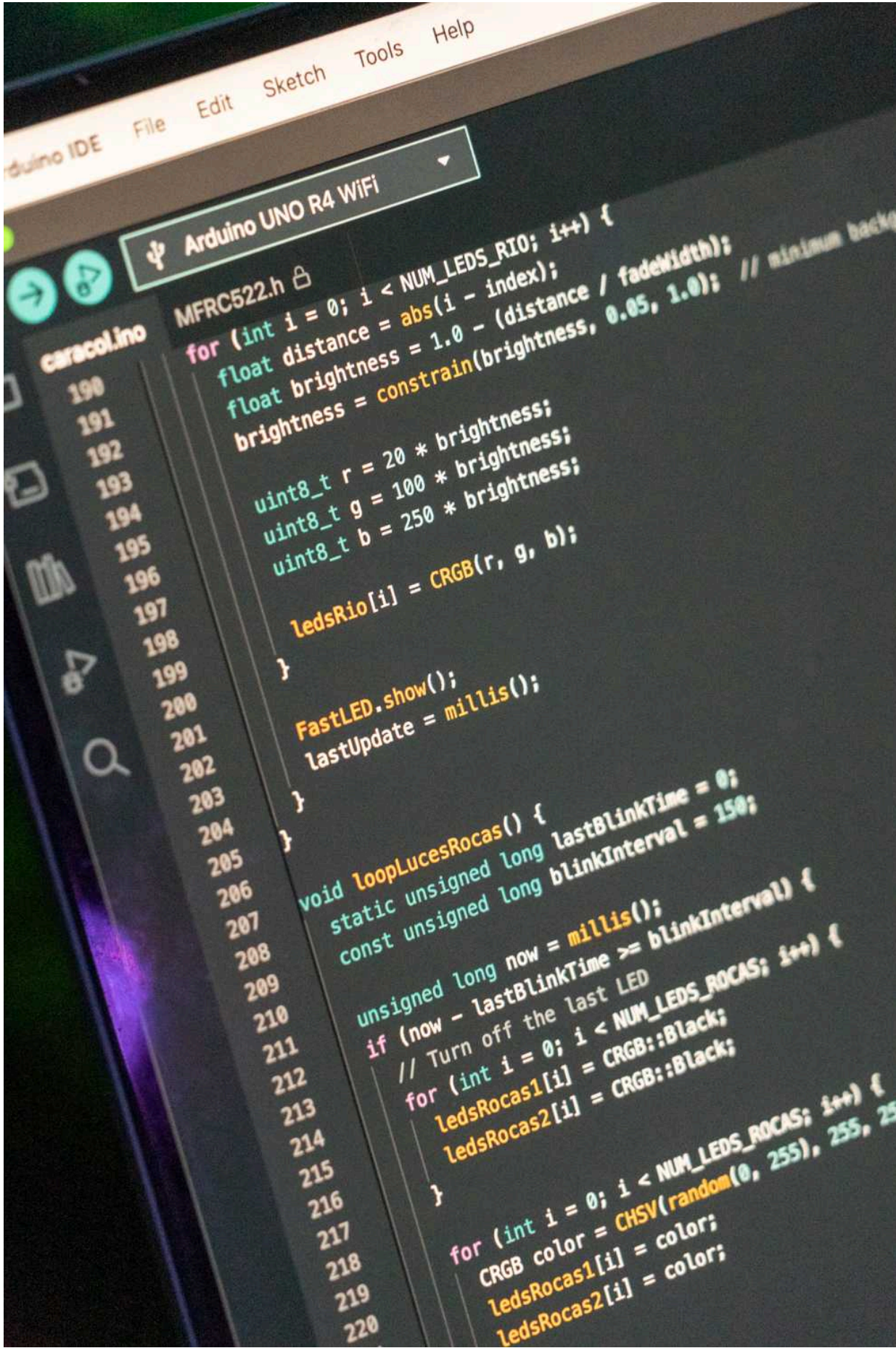
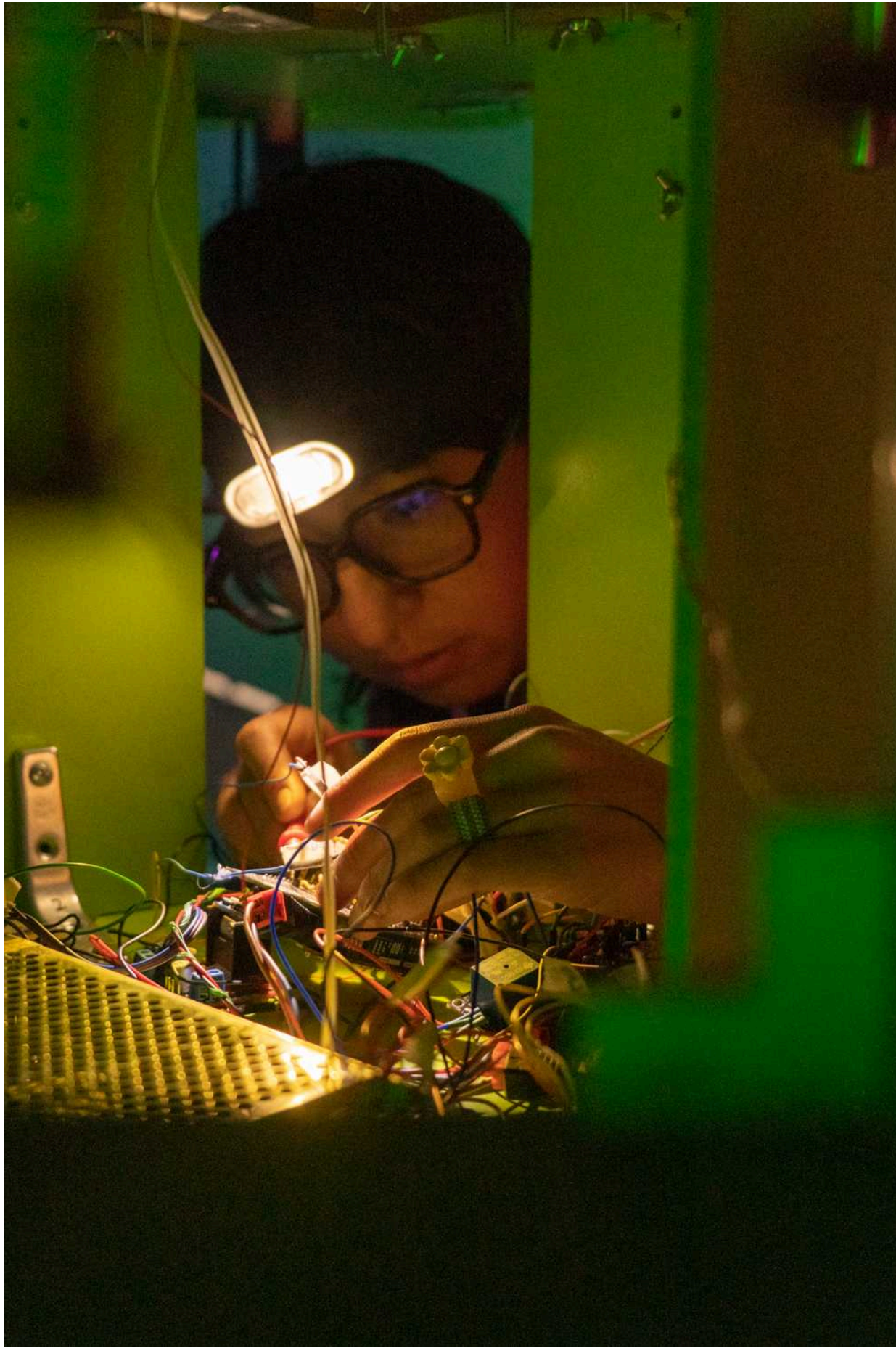
Flora generates a shared space for 2 or more people to work together and figure out what movements are necessary to bring everything to life.















Players decide whether they want to play freely or follow the provided fable and instructions: [https://flora\\_electronic\\_garden.surge.sh/](https://flora_electronic_garden.surge.sh/) (online mobile version)

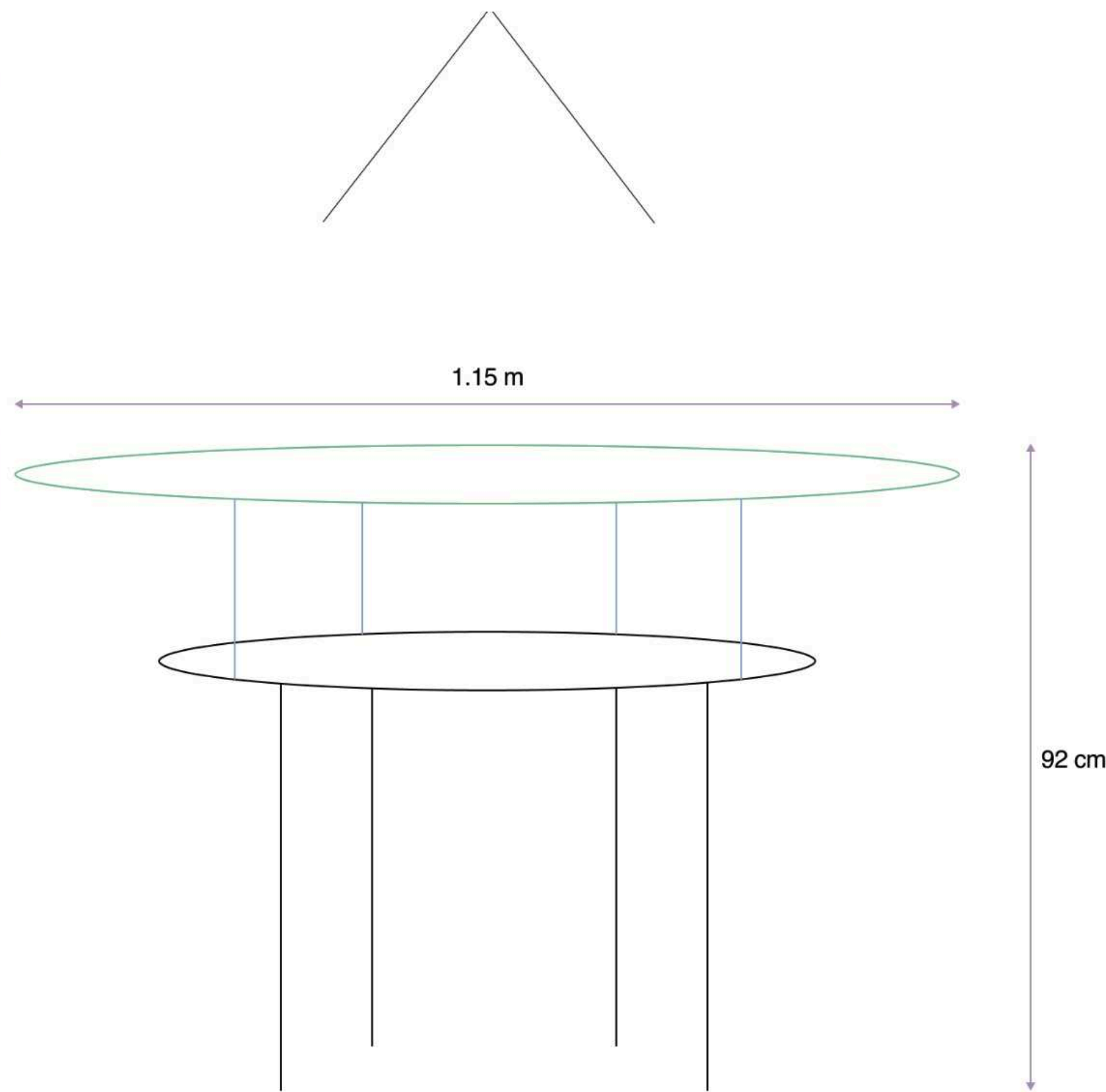
Space requirements for the installation

Projector

Game board

Electronics

Table



When setting up, we need a space with the following characteristics for installation:

- A minimum space of 3 cubic meters.
- Electrical outlets within a 1-meter radius of the board location.
- A structure above to hang a 3kg projector directly above the board.
- The ability to control all external light.

The following materials are also necessary:

- LED track light YWBL-WH
- 2 LED spotlights
- 2 power strips with 5 sockets (4 meters)
- 1 router



Levels of the game

|                    | Level 1                                      | Level 2                                                                   | Level 3                                                                | Level 4                                                                                   |                                                                                           | Level 5                                                                                                    |
|--------------------|----------------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
|                    | Seed, river and blooming flowers             | Snail, rocks and river                                                    | Lupine flowers and bees                                                | Carnivorous plants and flies                                                              |                                                                                           | Caterpillar and dancing flowers                                                                            |
| Action of players  | Plant the seed in the damp and fertile soil  | Assemble the puzzle of the snail's path and join the snail with its shell | Use the bee's mechanism to pass through each flower and pollinate them | Players have to choose between...                                                         |                                                                                           | Move the caterpillar from under the dead plant along the maze to help it exit on top of the center flowers |
|                    |                                              |                                                                           |                                                                        | Path A                                                                                    | Path B                                                                                    |                                                                                                            |
| Ecosystem response | The river activates and starts to flow water | The rocks light up, and water flows in the second part of the river       | The bee passes through each flower, and they change color              | Pick up the flies and feed them to the carnivorous plant A                                |                                                                                           | The flowers in the center of the board dance, and most of the ecosystem is alive.                          |
|                    |                                              |                                                                           |                                                                        | Carnivorous plant A is fed, alive and lit up.<br>Carnivorous plant B is dead and withers. | Carnivorous plant B is fed, alive and lit up.<br>Carnivorous plant A is dead and withers. |                                                                                                            |